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(54) **HYBRID ANTENNA STRUCTURE**

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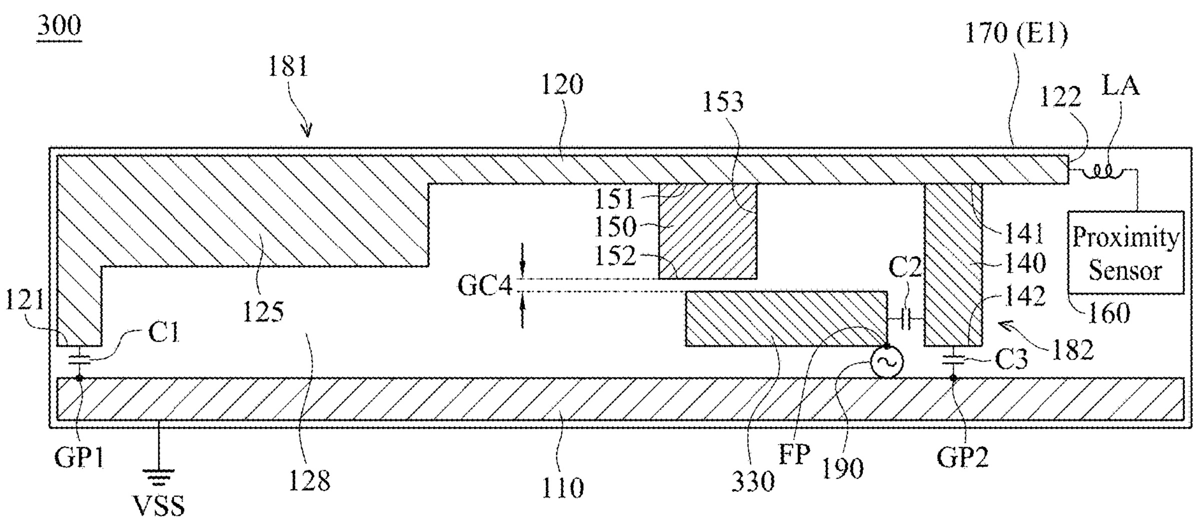
(57) **ABSTRACT**

A hybrid antenna structure includes a ground element, a main radiation element, a feeding radiation element, a shorting radiation element, an auxiliary radiation element, a proximity sensor, a dielectric substrate, an inductor, a first capacitor, a second capacitor, and a third capacitor. The main radiation element is coupled through the first capacitor to a first grounding point on the ground element. The feeding radiation element has a feeding point. The shorting radiation element is coupled to the main radiation element, and is coupled through the second capacitor to the feeding radiation element. The shorting radiation element is further coupled through the third capacitor to a second grounding point on the ground element. The auxiliary radiation element is coupled to the main radiation element. The auxiliary radiation element is adjacent to the feeding radiation element. The proximity sensor is coupled through the inductor to the main radiation element.



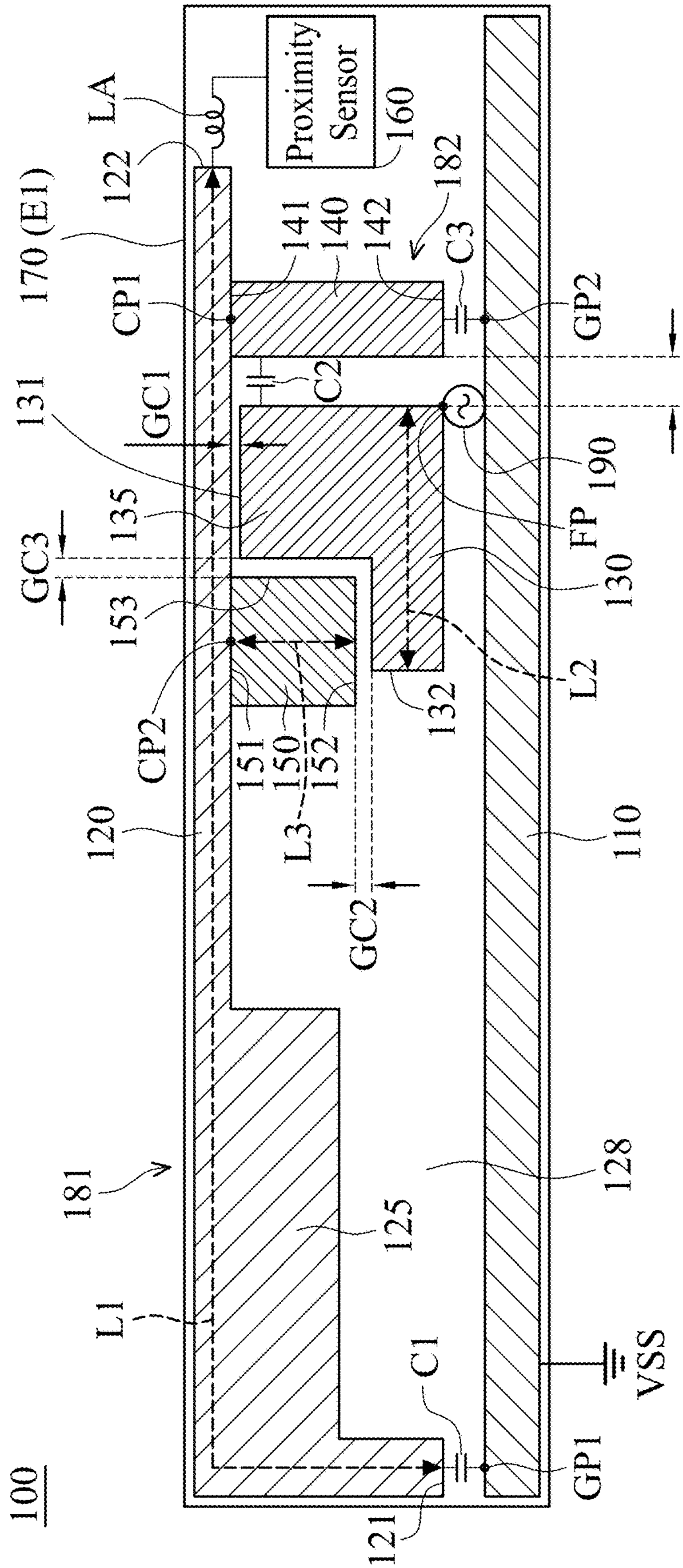

100
Proximity Sensor
160
LA
122
170 (E1)
182
141
140
142
C3
CP1
GP2
GC1
C2
D1
131
190
135
FP
GC3
130
153
L2
132
CP2
151
150
152
L3
120
GC2
110
181
128
125
L1
C1
VSS
GP1
121

FIG. 1

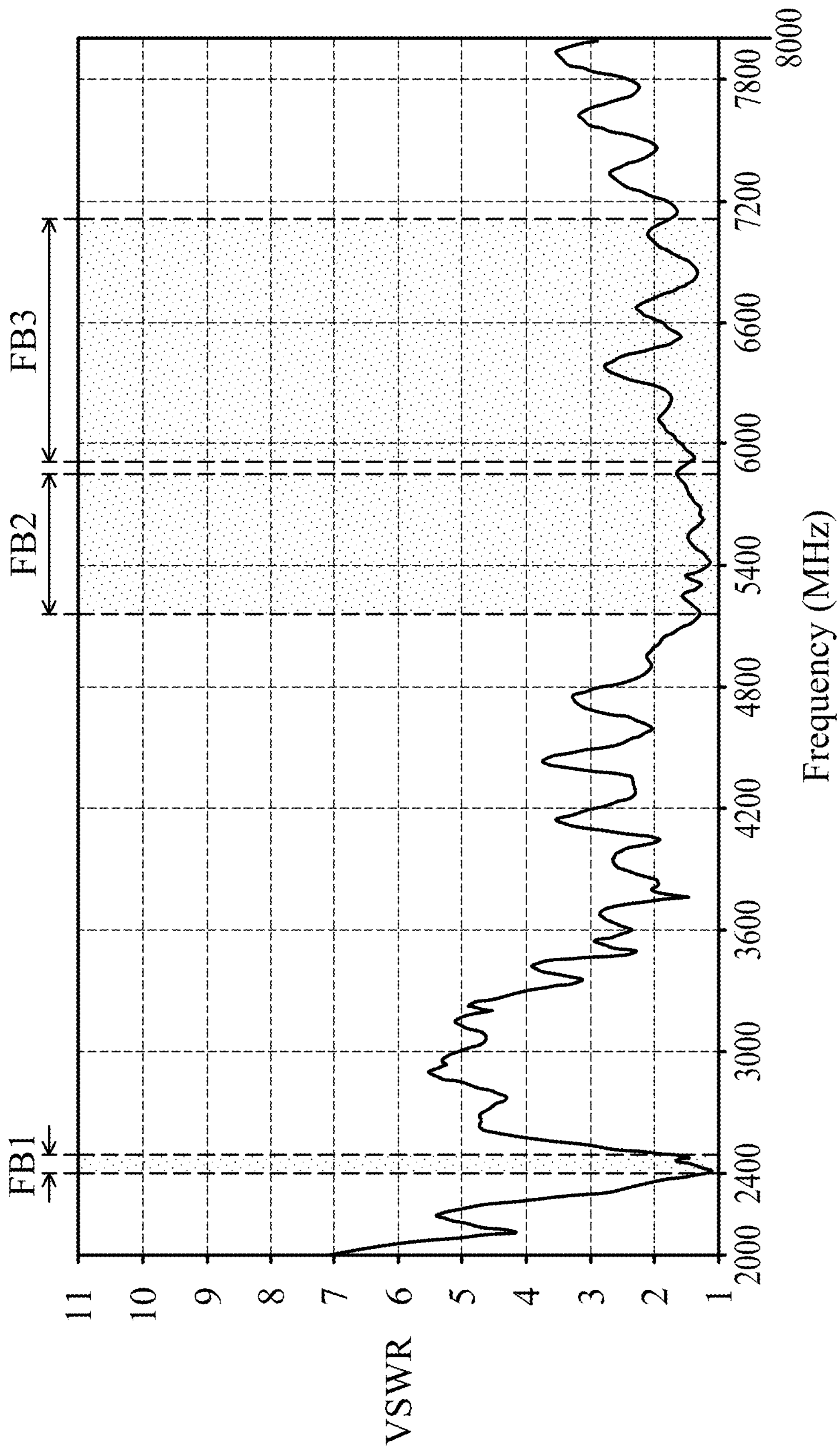

FB1
FB2
FB3
VSWR
11
10
9
8
7
6
5
4
3
2
1
2000
2400
3000
3600
4200
4800
5400
6000
6600
7200
7800
8000
Frequency (MHz)

FIG. 2

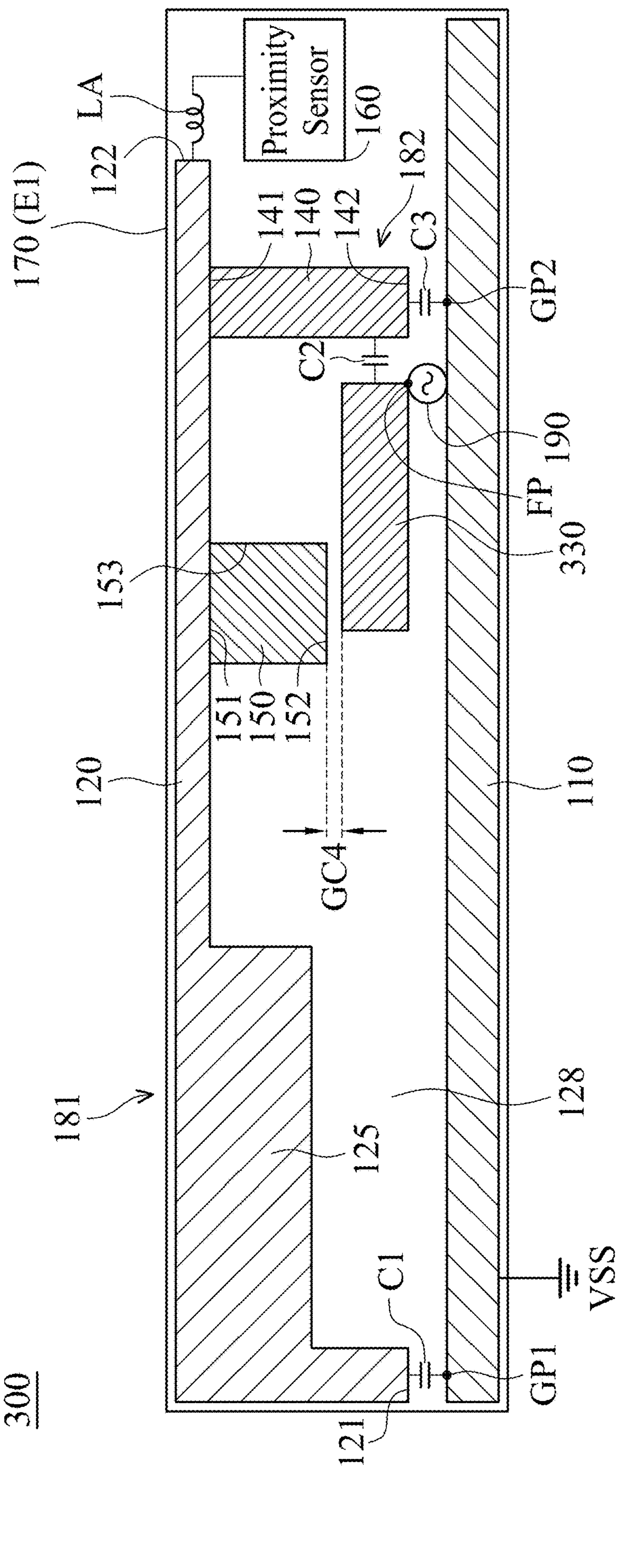

300
Proximity Sensor
160
LA
122
170 (E1)
141
140
142
182
C3
GP2
C2
190
FP
330
153
151
150
152
120
110
GC4
181
125
128
C1
VSS
GP1
121

FIG. 3

HYBRID ANTENNA STRUCTURE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority of Taiwan Patent Application No. 113108136 filed on Mar. 6, 2024, the entirety of which is incorporated by reference herein.

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

[0002] The disclosure generally relates to a hybrid antenna structure, and more particularly, to a wideband hybrid antenna structure.

Description of the Related Art

[0003] With the advancements being made in mobile communication technology, mobile devices such as portable computers, mobile phones, multimedia players, and other hybrid functional portable electronic devices have become more common. To satisfy consumer demand, mobile devices can usually perform wireless communication functions. Some devices cover a large wireless communication area; these include mobile phones using 2G, 3G, and LTE (Long Term Evolution) systems and using frequency bands of 700 MHZ, 850 MHz, 900 MHz, 1800 MHZ, 1900 MHZ, 2100 MHz, 2300 MHz, and 2500 MHz. Some devices cover a small wireless communication area; these include mobile phones using Wi-Fi systems and using frequency bands of 2.4 GHz, 5.2 GHz, and 5.8 GHz.

[0004] Antennas are indispensable elements for wireless communication. If an antenna used for signal reception and transmission has insufficient bandwidth, it will negatively affect the communication quality of the mobile device in which it is installed. Accordingly, it has become a critical challenge for antenna designers to design a small-size, wideband antenna element.

BRIEF SUMMARY OF THE DISCLOSURE

[0005] In an exemplary embodiment, the disclosure is directed to a hybrid antenna structure that includes a ground element, a main radiation element, a feeding radiation element, a shorting radiation element, an auxiliary radiation element, a proximity sensor, an inductor, a first capacitor, a second capacitor, and a third capacitor. The main radiation element is coupled through the first capacitor to a first grounding point on the ground element. The feeding radiation element has a feeding point. The shorting radiation element is coupled to the main radiation element, and is coupled through the second capacitor to the feeding radiation element. The shorting radiation element is further coupled through the third capacitor to a second grounding point on the ground element. The auxiliary radiation element is coupled to the main radiation element. The auxiliary radiation element is adjacent to the feeding radiation element. The proximity sensor is coupled through the inductor to the main radiation element.

BRIEF DESCRIPTION OF DRAWINGS

[0006] The disclosure can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:

[0007] FIG. **1** is a diagram of a hybrid antenna structure according to an embodiment of the disclosure;

[0008] FIG. **2** is a diagram of VSWR (Voltage Standing Wave Ratio) of a hybrid antenna structure according to an embodiment of the disclosure; and

[0009] FIG. **3** is a diagram of a hybrid antenna structure according to another embodiment of the disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0010] In order to illustrate the purposes, features and advantages of the disclosure, the embodiments and figures of the disclosure are shown in detail as follows.

[0011] Certain terms are used throughout the description and following claims to refer to particular components. As one skilled in the art will appreciate, manufacturers may refer to a component by different names. This document does not intend to distinguish between components that differ in name but not function. In the following description and in the claims, the terms "include" and "comprise" are used in an open-ended fashion, and thus should be interpreted to mean "include, but not limited to . . . ". The term "substantially" means the value is within an acceptable error range. One skilled in the art can solve the technical problem within a predetermined error range and achieve the proposed technical performance. Also, the term "couple" is intended to mean either an indirect or direct electrical connection. Accordingly, if one device is coupled to another device, that connection may be through a direct electrical connection, or through an indirect electrical connection via other devices and connections.

[0012] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed. Furthermore, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] FIG. **1** is a diagram of a hybrid antenna structure **100** according to an embodiment of the disclosure. The hybrid antenna structure **100** may be applied to a mobile device, such as a smart phone, a tablet computer, or a notebook computer. In the embodiment of FIG. **1**, the hybrid antenna structure **100** includes a ground element **110**, a main radiation element **120**, a feeding radiation element **130**, a shorting radiation element **140**, an auxiliary radiation element **150**, a proximity sensor **160**, a dielectric substrate **170**, an inductor LA, a first capacitor C**1**, a second capacitor C**2**, and a third capacitor C**3**. The ground element **110**, the main radiation element **120**, the feeding radiation element **130**, the shorting radiation element **140**, and the auxiliary radiation element **150** may all be made of metal materials, such as copper, silver, aluminum, iron, or their alloys.

[0014] The ground element **110** is configured to provide a ground voltage VSS. For example, the ground element **110** may be implemented with a ground copper foil. In some embodiments, the ground element **110** is further coupled to a system ground plane (not shown) of the hybrid antenna structure **100**. In addition, there are a first grounding point GP**1** and a second grounding point GP**2** on the ground element **110**. The first grounding point GP**1** and the second grounding point GP**2** may be different from each other.

[0015] The main radiation element **120** is coupled through the first capacitor C**1** to the first grounding point GP**1** on the ground element **110**. Specifically, the main radiation element **120** has a first end **121** and a second end **122**. A terminal of the first capacitor C**1** is coupled to the first end **121** of the main radiation element **120**, and another terminal of the first capacitor C**1** is coupled to the first grounding point GP**1**. In some embodiments, the main radiation element **120** substantially has a relatively large L-shape, but it is not limited thereto. In some embodiments, the main radiation element **120** further includes a bending and widening portion **125**, which may substantially have a rectangular shape.

[0016] The feeding radiation element **130** has a first end **131** and a second end **132**. A feeding point FP may be positioned at a corner of the feeding radiation element **130**. The second end **132** of the feeding radiation element **130** is an open end. The feeding point FP may be further coupled to a signal source **190**. For example, the signal source **190** may be an RF (Radio Frequency) module for exciting the hybrid antenna structure **100**. In some embodiments, the feeding radiation element **130** further includes an extension portion **135** positioned at the first end **131**, and the extension portion **135** substantially has another rectangular shape. Because of the aforementioned extension portion **135**, a coupling gap GC**1** may be formed between the first end **131** of the feeding radiation element **130** and the main radiation element **120**. In some embodiments, the feeding radiation element **130** substantially has a relatively small L-shape (compared with the main radiation element **120**), but it is not limited thereto. It should be noted that the term "adjacent" or "close" over the disclosure means that the distance (spacing) between two corresponding elements is smaller than a predetermined distance (e.g., 10 mm or the shorter), but often does not mean that the two corresponding elements directly touch each other (i.e., the aforementioned distance/spacing between them is reduced to 0).

[0017] The shorting radiation element **140** is coupled through the second capacitor C**2** to the feeding radiation element **130**. Specifically, the shorting radiation element **140** has a first end **141** and a second end **142**. The first end **141** of the shorting radiation element **140** is coupled to a first connection point CP**1** on the main radiation element **120**. A terminal of the second capacitor C**2** is coupled to the first end **131** of the feeding radiation element **130**, and another terminal of the second capacitor C**2** is coupled to the first end **141** of the shorting radiation element **140**. In addition, the shorting radiation element **140** is further coupled through the third capacitor C**3** to the second grounding point GP**2** on the ground element **110**. That is, a terminal of the third capacitor C**3** is coupled to the second end **142** of the shorting radiation element **140**, and another terminal of the third capacitor C**3** is coupled to the second grounding point GP**2**. In some embodiments, the shorting radiation element **140** substantially has a straight-line shape, but it not limited thereto.

[0018] The auxiliary radiation element **150** is coupled to the main radiation element **120**. Specifically, the auxiliary radiation element **150** has a first end **151**, a second end **152**, and a side **153**. The first end **151** of the auxiliary radiation element **150** is coupled to a second connection point CP**2** on the main radiation element **120**. The second end **152** of the auxiliary radiation element **150** is an open end. The second connection point CP**2** may be different from the first connection point CP**1**. The auxiliary radiation element **150** is adjacent to the feeding radiation element **130**. For example, a coupling gap GC**2** may be formed between the second end **152** of the auxiliary radiation element **150** and the feeding radiation element **130**, and another coupling gap GC**3** may be formed between the side **153** of the auxiliary radiation element **150** and the feeding radiation element **130**. In some embodiments, the auxiliary radiation element **150** substantially has a rectangular shape or a square shape, but it is not limited thereto.

[0019] In some embodiments, the feeding radiation element **130**, the shorting radiation element **140**, and the auxiliary radiation element **150** are at least partially surrounded by the main radiation element **120**. In alternative embodiments, a slot region **128** is defined between the ground element **110** and the main radiation element **120**. The feeding radiation element **130**, the shorting radiation element **140**, and the auxiliary radiation element **150** may all be disposed inside the slot region **128**.

[0020] The proximity sensor **160** is coupled through the inductor LA to the main radiation element **120**. For example, a terminal of the inductor LA may be coupled to the second end **122** of the main radiation element **120**, and another terminal of the inductor LA may be coupled to the proximity sensor **160**. However, the disclosure is not limited thereto. In alternative embodiments, the proximity sensor **160** is coupled through the inductor LA to any other position on the main radiation element **120**, without affecting the performance thereof. Generally, the main radiation element **120** is configured as a sensing pad of the proximity sensor **160**, thereby increasing the detectable distance of the proximity sensor **160**.

[0021] The dielectric substrate **170** may be an FR4 (Flame Retardant 4) substrate, a PCB (Printed Circuit Board), or an FPC (Flexible Printed Circuit). In some embodiments, the ground element **110**, the main radiation element **120**, the feeding radiation element **130**, the shorting radiation element **140**, and the auxiliary radiation element **150** are all disposed on the same surface E**1** of the dielectric substrate **170**. In other words, the hybrid antenna structure **100** belongs to a planar antenna structure, so as to reduce the overall manufacturing cost. In alternative embodiments, the proximity sensor **160**, the inductor LA, the first capacitor C**1**, the second capacitor C**2**, and the third capacitor C**3** are all disposed on the aforementioned surface E**1** of the dielectric substrate **170**, but they are not limited thereto.

[0022] FIG. **2** is a diagram of VSWR (Voltage Standing Wave Ratio) of the hybrid antenna structure **100** according to an embodiment of the disclosure. The horizontal axis represents the operational frequency (MHz), and the vertical axis represents the VSWR. According to the measurement of FIG. **2**, the hybrid antenna structure **100** can cover a first frequency band FB**1**, a second frequency band FB**2**, and a third frequency band FB**3**. For example, the first frequency band FB**1** may be from 2400 MHz to 2500 MHz, the second frequency band FB**2** may be from 5150 MHz to 5850 MHz, and the third frequency band FB**3** may be from 5925 MHz to 7125 MHz. Therefore, the hybrid antenna structure **100** can support the wideband operations of WLAN (Wireless Local Area Networks), Wi-Fi **6**E, and Wi-Fi **7**.

[0023] The operational principles of the hybrid antenna structure **100** of some embodiments are described as follows: An outer loop structure **181** is formed by the feeding radiation element **130**, the second capacitor C**2**, the shorting radiation element **140**, the main radiation element **120**, and the first capacitor C**1**. The outer loop structure **181** can be excited to generate the first frequency band FB**1**. An inner loop structure **182** is formed by the feeding radiation element **130**, the second capacitor C**2**, the shorting radiation element **140**, and the third capacitor C**3**. The inner loop structure **182** can be excited to generate the second frequency band FB**2** and the third frequency band FB**3**. The bending and widening portion **125** of the main radiation element **120** is configured to fine-tune the impedance matching of the first frequency band FB**1**. The auxiliary radiation element **150** is configured to fine-tune the impedance matching of the second frequency band FB**2** and the third frequency band FB**3**.

[0024] According to practical measurements, the inductor LA can prevent alternating currents of the hybrid antenna structure **100** from entering the proximity sensor **160**. Furthermore, the first capacitor C**1**, the second capacitor C**2**, and the third capacitor C**3** can prevent direct current from the proximity sensor **160** from entering the ground element **110**. With the proposed design, because the main radiation element **120** used as the sensing pad is well integrated, the hybrid antenna structure **100** can provide the functions of both wideband operation and proximity sense, without additionally increasing the overall device size. It should be also understood that the connection position of the second capacitor C**2** is adjustable between the first end **131** and the second end **132** of the feeding radiation element **130**. If the connection position of the second capacitor C**2** is different, the effective resonant length of the inner loop structure **182** can also be changed, so as to appropriately adjust the frequency shift of the second frequency band FB**2** and the third frequency band FB**3**.

[0025] The element sizes and parameters of the hybrid antenna structure **100** of some embodiments are described as follows: The length L**1** of the main radiation element **120** may be from 0.25 to 0.5 wavelength (0.25π~0.5λ) of the first frequency band FB**1** of the hybrid antenna structure **100**, such as about 0.4 wavelength (0.4λ). The length L**2** of the feeding radiation element **130** may be from 0.25 to 0.5 wavelength (0.25λ~0.5λ) of the second frequency band FB**2** or the third frequency band FB**3** of the hybrid antenna structure **100**, such as about 0.3 wavelength (0.32). The length L**3** of the auxiliary radiation element **140** may be from 0.5 mm to 2.5 mm. The width of the coupling gap GC**1** may be form 0.2 mm to 3.5 mm. The width of the coupling gap GC**2** may be form 0.5 mm to 1.5 mm. The width of the coupling gap GC**3** may be form 0.5 mm to 1.5 mm. The distance D**1** between the shorting radiation element **140** and the feeding radiation element **130** may be from 0.2 mm to 2 mm.

[0026] The capacitance of the first capacitor C**1** may be greater than or equal to 10 pF. The capacitance of the second capacitor C**2** may be greater than or equal to 10 pF. The capacitance of the third capacitor C**3** may be greater than or equal to 27 pF. The inductance of the inductor LA may be greater than or equal to 30 nH. The above ranges of element sizes and parameters are calculated and obtained according to many experimental results, and they help to optimize the operational bandwidth and the impedance matching of the hybrid antenna structure **100**, and also to maximize the detectable distance of the proximity sensor **160**.

[0027] The following embodiments will introduce different configurations and detailed structural features of the hybrid antenna structure **100**. It should be understood that these figures and descriptions are merely exemplary, rather than limitations of the disclosure.

[0028] FIG. **3** is a diagram of a hybrid antenna structure **300** according to another embodiment of the disclosure. FIG. **3** is similar to FIG. **1**. In the embodiment of FIG. **3**, a feeding radiation element **330** of the hybrid antenna structure **300** substantially has a simple straight-line shape, and the connection position of the second capacitor C**2** is also changed. Since the feeding radiation element **330** does not include any extension portion, there can be only one coupling gap GC**4** formed between the auxiliary radiation element **150** and the feeding radiation element **330**. According to practical measurements, this design can help to further adjust the impedance matching of the second frequency band FB**2** and the third frequency band FB**3** of the hybrid antenna structure **300**. Other features of the hybrid antenna structure **300** of FIG. **3** are similar to those of hybrid antenna structure **100** of FIG. **1**. Accordingly, the two embodiments can achieve similar levels of performance.

[0029] The disclosure proposes a novel hybrid antenna structure. In comparison to the conventional design, the disclosure has the advantages of small size, wide bandwidth, proximity sense, high communication quality, and low manufacturing cost. Therefore, the disclosure is suitable for applications in a variety of mobile communication devices.

[0030] Note that the above element sizes, element shapes, element parameters, and frequency ranges are not limitations of the disclosure. An antenna designer can fine-tune these settings or values in order to meet specific requirements. It should be understood that the hybrid antenna structure of the disclosure is not limited to the configurations depicted in FIGS. **1-3**. The disclosure may merely include any one or more features of any one or more embodiments of FIGS. **1-3**. In other words, not all of the features displayed in the figures should be implemented in the hybrid antenna structure of the disclosure.

[0031] Use of ordinal terms such as “first”, “second”, “third”, etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed, but are used merely as labels to distinguish one claim element having a certain name from another element having the same name (but for use of the ordinal term) to distinguish the claim elements.

[0032] While the disclosure has been described by way of example and in terms of the preferred embodiments, it should be understood that the disclosure is not limited to the disclosed embodiments. On the contrary, it is intended to cover various modifications and similar arrangements (as would be apparent to those skilled in the art). Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

What is claimed is:

1. A hybrid antenna structure, comprising:
- a ground element;
- a main radiation element;
- a first capacitor, wherein the main radiation element is coupled through the first capacitor to a first grounding point on the ground element;
- a feeding radiation element, having a feeding point;
- a second capacitor;
- a shorting radiation element, coupled to the main radiation element, and coupled through the second capacitor to the feeding radiation element;
- a third capacitor, wherein the shorting radiation element is further coupled through the third capacitor to a second grounding point on the ground element;
- an auxiliary radiation element, coupled to the main radiation element, wherein the auxiliary radiation element is adjacent to the feeding radiation element;
- an inductor; and
- a proximity sensor, coupled through the inductor to the main radiation element.

2. The hybrid antenna structure as claimed in claim **1**, wherein the main radiation element is configured as a sensing pad of the proximity sensor.

3. The hybrid antenna structure as claimed in claim **1**, further comprising:
- a dielectric substrate, wherein the ground element, the main radiation element, the feeding radiation element, the shorting radiation element, and the auxiliary radiation element are disposed on the dielectric substrate.

4. The hybrid antenna structure as claimed in claim **1**, wherein the main radiation element substantially has a large L-shape.

5. The hybrid antenna structure as claimed in claim **1**, wherein the main radiation element further comprises a bending and widening portion.

6. The hybrid antenna structure as claimed in claim **1**, wherein the feeding radiation element substantially has a small L-shape.

7. The hybrid antenna structure as claimed in claim **1**, wherein the feeding radiation element substantially has a straight-line shape.

8. The hybrid antenna structure as claimed in claim **1**, wherein the auxiliary radiation element substantially has a rectangular shape or a square shape.

9. The hybrid antenna structure as claimed in claim **1**, wherein a coupling gap is formed between the auxiliary radiation element and the feeding radiation element.

10. The hybrid antenna structure as claimed in claim **1**, wherein the hybrid antenna structure covers a first frequency band, a second frequency band, and a third frequency band.

11. The hybrid antenna structure as claimed in claim **10**, wherein the first frequency band is from 2400 MHz to 2500 MHz, the second frequency band is from 5150 MHz to 5850 MHz, and the third frequency band is from 5925 MHz to 7125 MHz.

12. The hybrid antenna structure as claimed in claim **10**, wherein a length of the main radiation element is from 0.25 to 0.5 wavelength of the first frequency band.

13. The hybrid antenna structure as claimed in claim **10**, wherein a length of the feeding radiation element is from 0.25 to 0.5 wavelength of the second frequency band or the third frequency band.

14. The hybrid antenna structure as claimed in claim **10**, wherein an outer loop structure is formed by the feeding radiation element, the second capacitor, the shorting radiation element, the main radiation element, and the first capacitor.

15. The hybrid antenna structure as claimed in claim **14**, wherein the outer loop structure is excited to generate the first frequency band.

16. The hybrid antenna structure as claimed in claim **1**, wherein an inner loop structure is formed by the feeding radiation element, the second capacitor, the shorting radiation element, and the third capacitor.

17. The hybrid antenna structure as claimed in claim **16**, wherein the inner loop structure is excited to generate the second frequency band and the third frequency band.

18. The hybrid antenna structure as claimed in claim **1**, wherein a capacitance of each of the first capacitor and the second capacitor is greater than or equal to 10 pF.

19. The hybrid antenna structure as claimed in claim **1**, wherein a capacitance of the third capacitor is greater than or equal to 27 pF.

20. The hybrid antenna structure as claimed in claim **1**, wherein an inductance of the inductor is greater than or equal to 30 nH.

* * * * *